

Battery Swapping Mobile Service Station for Electric Vehicles Application

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Abstract—This paper provides a novel approach for providing battery swapping services to electric vehicle users, centered around a Battery swapping mobile station. Firstly, this paper establishes the structural architecture for the smooth and efficient to provide swapping services to users. Also establishes the functions and roles of the various entities present in the organization and defines the difference between charging stations and swapping stations. Secondly, gives a brief description of the working of the mechanism involved in the mobile swapping unit. Also, the mobile application and the communication between the user and the service management system and establishes how the user can send requests, raise complaints through the application. Thirdly, the power consumed by the battery swapping unit, its run time, and how the solar-assisted charging is done. The above is backed by mathematical analysis.

Index Terms—Electric vehicle; Battery swapping; Battery swapping station; Android application

I. INTRODUCTION

In recent years, there has been a significant increase in the adoption of electric vehicles as the main mode of transportation. The key issue for the effective operation of electrical vehicle (EV) is energy replenishment. There are two solutions for this issue, they are charging stations and swapping stations. The charging stations take a long time to charge the EV battery, which causes range anxiety, wait-anxiety among the users. Installation of the charging station requires a large amount of energy, land acquisition to keep up with the demand, which leads to an increase in expenses, Another solution for this issues is battery swapping service. The battery swapping stations have the advantage of quicker change time, but also require large space to swap multiple batteries at a time. However, adding the mobile swapping stations to the mix with swapping stations would make the process much effective and efficient. The mobile station is tasked to arrive at the user's preferred location, and perform swapping operations within minutes. With the help of mobile app and service management, the services would be much smoother. Depending on the location accessibility and roads the efficiency of services varies [1]–[6]. Objectives of the proposed work framed as:

- Establishing an EVs battery swapping service architecture centered around battery swapping mobile station.
- To provide efficient and effective swapping services.

- Establish an idea of a battery swapping mechanism compact enough to fit in the back of a truck.
- Mobile to have smooth two-way communication between user and service provider.

In the last decades, there has been a rapid increase in the work related to electric vehicles. As EVs are considered as the future. Many studies have been looking into environmental impact, limitations of adoption, and its impact on the electrical grid. Most of the research indicates that the primary issue with the adoption of EVs is energy replenishment. The energy replenishment solution is charging stations and swapping stations, and previous research establishes the site selections, replenishment strategies, their constraints, and so on. Like optimal BSS placement reduces the impacts to distribution system performance and analytical method to give results similar to ABC for optimal BSS placement [7]–[11]. Several studies have been investigated for the optimal business model for swapping services [12]–[16]. Most of the research assumes that the service to be static, but our approach is more dynamic as the mobile station travels around to provide services, from the published literature some authors exchange process works, and provide an efficient station assignment. It is also observing that some research articles provides mathematical proof of how the approach impacts the air quality and human health conditions [17]–[23], [25]. Therefore, considering the previous research the mobile stations has can be a unique addition to the EV energy replenishment strategies .

II. METHODOLOGY

Methodology of the work is divided in two part as a. Battery swapping service architecture and battery swapping mobile station.

A. Battery swapping service architecture

The adaption of battery swapping mobile stations comes with the establishment of several entities. Every participant has been assigned a specific task, such that participation and constant communication between them is necessary for smooth and efficient battery swapping services. These roles include battery manufacturing, charging, storing, swapping, transportation and management. The flow of the process between them is shown in Fig.1. The manufacturers are a crucial part of the

whole organization, as it the first step in the architecture. They provide fresh standardized batteries to the service provider. As there are many electric vehicle manufacturers each with a unique battery design, such as different weight, size, performance, etc. This creates a problem of designing specific battery swapping mechanisms for different EV manufacturers, which is neither efficient nor cost-effective. Hence, having a standardized battery design for all EV's would be more efficient and cost-effective. With this commonality of battery design, we can have the same swapping mechanism for several different EVs.

As there is no charging element present in the swapping van the batteries have to be charged elsewhere. Thus, a centralized charging hub called a Battery charging station is established. The batteries are charged and sent out to the battery swapping stations (local or mobile stations) to be swapped out for the user's low charge battery. This can also double as a warehouse for stockpiling batteries for when there is a spike in demand for batteries. The centralized battery charging requires huge amounts of energy, hence the ideal location for its installation would be near a substation or renewable energy plants such as solar farms, wind farms, etc. For efficient operation of the station, it would be optimal to utilize energy from the smart grid during off-peak and using renewable energy sources during peak hours. And also provides a suitable architecture for the implementation of V2G technology. As the battery swapping mobile station is confined to carrying a confined amount of batteries, it is not possible to deliver batteries to the users at an efficient rate. So the mobile swapping station has to get its fully charged batteries from the charging stations, which would not be efficient as it's far from the inner city. To guarantee battery replenishment, a local battery-swapping station is deployed with a battery swapping mechanism, charging systems, and a stockpile of batteries. Once the mobile station has completely exhausted fully charged batteries, it can get replenish its supply from the local station and provide uninterrupted services to the users. It also has a battery swapping mechanism installed in it, if the user is nearby they can use the local station to get a fully charged battery as it would be inefficient to wait for the mobile station. During the peak commute hours, sometimes the supply of fully charged batteries would be insufficient, and waiting for the charging station to send in batteries would delay the services, thus, a battery charging system is present in it to keep up with the demand. To reduce the stress on the local station, the service area is divided into zones based on the population density, users registered, and road density. Each zone is shared between two local and mobile stations, covering one-half of the zone. And are installed at adjacent corners of the zone. For example, let us consider a zone of 5 sq km, it's divided diagonally and shared among them. As charging is involved, a preferable location would be near a distribution substation or distribution transformer. It is the most important among the six entities present in the swapping architecture, as the main aim of this entity is to overlook the operation and function of every other entity. Also establishes constant communication between

the entities so that the operation of the swapping services is smooth and efficient. The service management systems manage the records of the users, the service registrations, real-time pricing, location of the mobile station, status of the mobile station, and the energy consumption of the charging factory, etc. One of the most important tasks of this entity is overlooking the swapping process when the user sends a request for a swap, they take care of scheduling and assigning the task to a nearby mobile station. Thus, they have to keep track of the location, battery status, driving direction. After the swapping is complete they have to keep records of the state of charge of the battery swapped and how many swaps has the user completed, to decide the pricing of the services offered. And also have take care of complaints through customer care services. They provide the above details to the user via the mobile application.

B. Battery swapping mobile station

The battery swapping mobile station is equipped with an automatic positioning system and a communication system that can communicate with the battery swapping service management system and EV users via wireless communication technology. The communication unit sends the real-time location, direction, speed, full charge battery available in the mobile station to the management system. With this information, whenever a user requests the services, scheduling and the response is communicated to the user through the mobile app. Battery swapping organizational structure has shown in Fig.3.

The whole mechanism can be divided into three key functioning parts (or mechanisms) namely

- 1) Swapping unit
- 2) Re-positioning unit
- 3) Communication unit

Swapping unit- The function of this mechanism is to swap out the low charge battery on the EV for a fully charged battery mobile station. The batteries are placed in a predestined position to make the process much faster, as it always grabs the battery from its default rack. The car enters the mobile station through position tracks, once the car is in proper position the swapping mechanism elevates with the help of threaded rods coupled to a stepper motor. It learns the position of the car with the help of limit switches, position sensors. And the screwing mechanism is deployed, this unscrews the battery and pushes it into the designated rack through the conveyor. And grabs the fully charged battery and uses the same process backward to install it.

Re-positioning unit- As discussed in the previous section the batteries have to be positioned in a defined rack in order to perform swapping operation. The purpose of this unit is to re-rack the batteries into position after every swap is completed. It is equipped with sensors to know the battery's presence, limit switches to know if the machine is elevated to the level of the rack. The rack where the batteries are kept is equipped with a conveyor to push the batteries in and out of them. The racks are static, so the swapping mechanism can be raised and

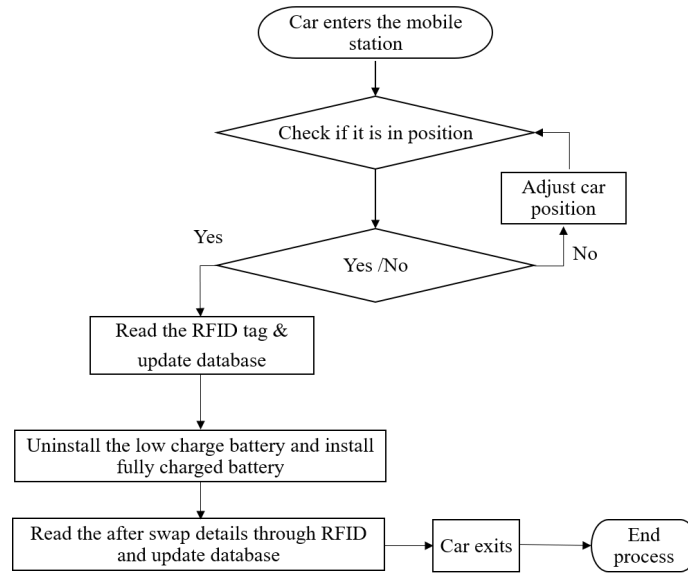


Fig. 1. Battery swapping organizational structure.

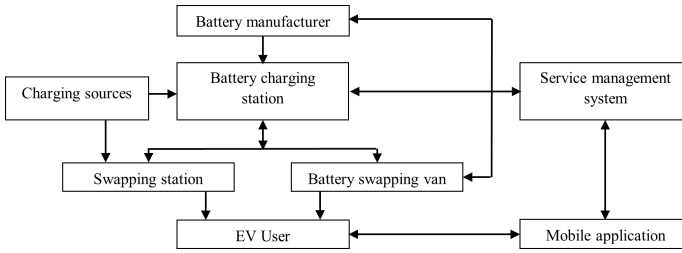


Fig. 2. Battery swapping formulation.

lowered hence it is used to move the batteries from one rack to other.

Fig.4 describes the battery's predefined position. After the swap is performed the battery position is changed to one shown in figure 1b. The procedure to get from 1b to 1c requires the swapping mechanism to collect the battery from the top and place it at the bottom. Then collect the low voltage battery from the bottom second rack and place it at the top. And after the subsequent swap, the operation is repeated but instead of the top, it collects the next fully charged battery and so on.

Communication unit- As established earlier the van has to be in constant communication with the service management system, the communication unit comes into play here. It consists of a GSM module or WIFI router to send data wireless, but the GSM module is preferred here because we need not worry about losing contact as long as there is good reception. GPRS tracker to know the location, direction in which it is headed. It also has an RFID reader to get data about the battery installed and its performance, state of charge. It reads to the tag twice once before swapping and once after swapping. And sends the data to management.

III. RESULTS AND DISCUSSION

A. Mobile application response

The mobile application is the point of contact between the user and the service provider, it is managed by the service management system. The user can install the mobile application on their smartphone to send service requests, real-time location, state of charge, request status. In addition, also provides real-time information exchange between user and provider. Moreover, the user can also raise complaints, gather information through the app. Mobile Applications elements and their functions are explained in Fig 6. Every user is provided with a unique login id and a password so that only the user can gain access to his/her profile. The user can fill the login credentials in the space provided as shown in the Fig.6. Request page - Through this page, the user can send a request to avail battery swapping service. The user can select the location to which the mobile station has to be sent, by moving the marker on the map. After selecting the location press the request button to send a request. The checkbox indicates the request status whether it's accepted or no request made. By any chance, the user wants to cancel the request he can do it by clicking the cancel button. If the request is accepted, he can track it by clicking the track button which indicates the real-time location of the mobile station.

Car background-This page displays all the essentials details, such as swap id, the battery that is installed after the swap, swap time, and the installed battery performance.

Battery performance- This page shows the performance of the battery that is currently installed in the EV, how many times its been cycled, and the time when it was swapped.

B. Solar array for an assisted charge

The operation of the mobile station requires a power source. So, an auxiliary battery for powering the mechanisms. And

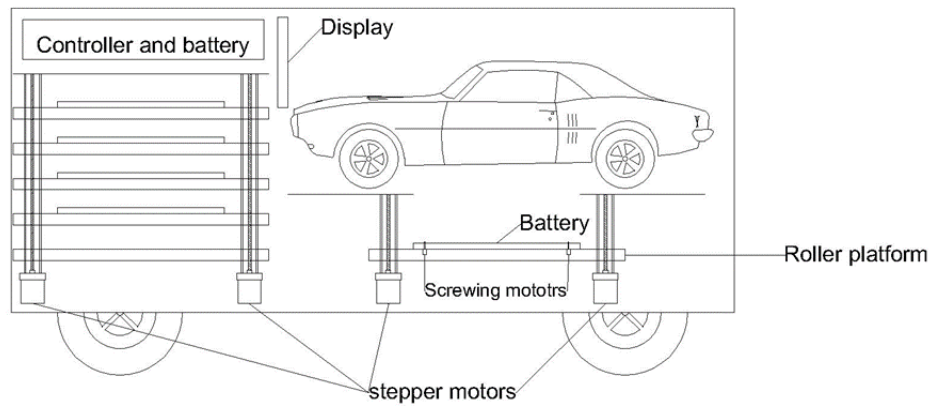


Fig. 3. Battery swapping organizational structure.

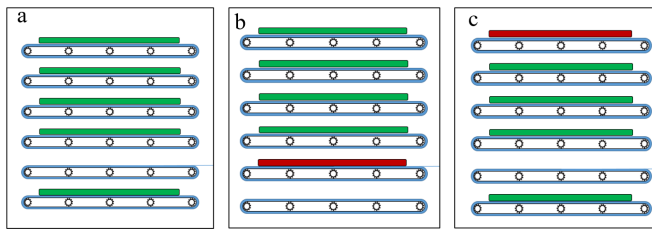


Fig. 4. Battery swapping mechanism.

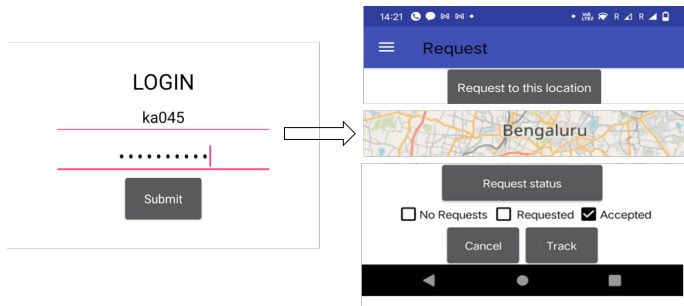


Fig. 5. Android application snapshot1.

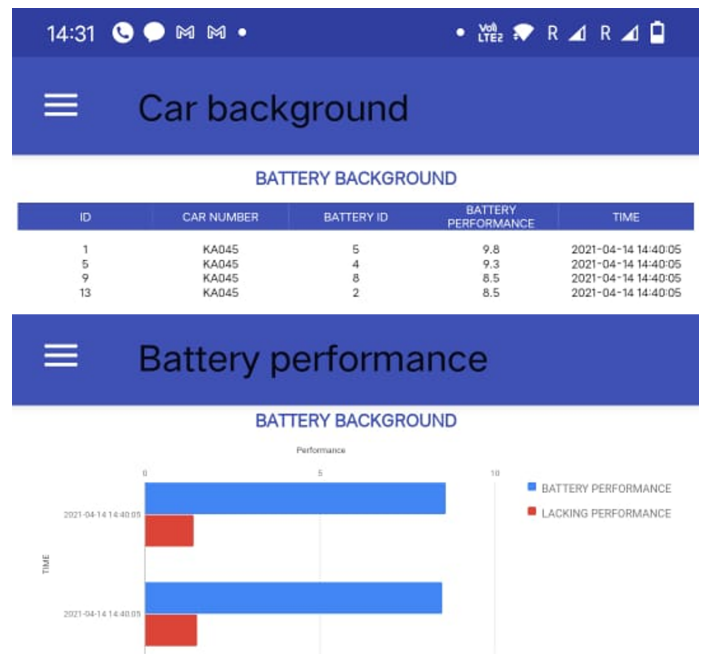


Fig. 6. Android application snapshot2.

this battery also requires charging, and the roof mobile station has a lot of space wasted. This space can be used to install a solar panel and use the energy from the panels can be used to charge the auxiliary battery in the mobile station. Let us consider 4 battery modules for the auxiliary battery module. And each module consists of 5 kWh a nominal voltage of 22 V. And they are connected in a series-parallel connection because if one battery is damaged it would still function. The battery connections are made as shown in Fig 7. There are two parallel connections and two series connections. For parallel connections, the voltage stays constant, for series connections the current stays constant.

For 4 modules the capacity: $4 \times 5 = 20$ kWh

For series nominal voltage: $22 \times 2 = 44$ V

For charging a battery the charging voltage should be more than the nominal voltage.

TABLE I
SOLAR ARRAY SPECIFICATIONS

Dimension	1956 × 992 × 36 mm
Max output current per MPPT (A)	9.63 A
Maximum output power	370 Watt
Maximum output Voltage (V)	48 V

The panels are in a series-parallel connection as shown in Fig.7, batteries are connected in series and then connected in parallel.

For the 2 series connections - $2 \times 48 = 96$ V

For the 3 parallel connections - $3 \times 9.63 = 28.89$ A

For 6 panels power output - $6 \times 370 = 2,220$ W

Approximate time taken to charge the modules, considering

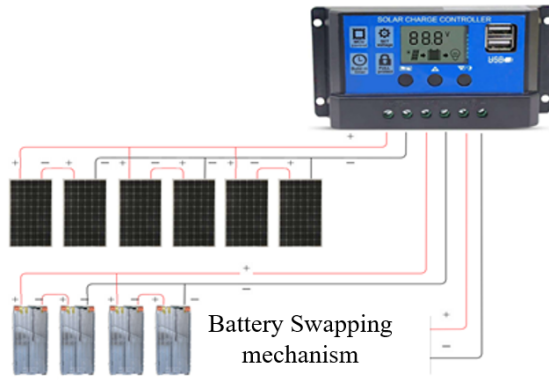


Fig. 7. Battery module.

the max conditions:

$$Time = \frac{Capacity}{Inputpower} \quad (1)$$

$$\frac{20kWh}{2220W} = 9h \quad (2)$$

C. Power consumption and run time

This section describe the several components like a PLC controller, stepper motor, brush-less DC motors, limit switches, proximity sensors, sensors, etc. This section gives us an approximate run time and power consumption of the mobile station.

$$Runtimeavailable = \frac{Totalcapacity(kWh)}{load(kW)} \quad (3)$$

$$= 20 \text{ KWh} / 2090 \text{ W} = 9.5 \text{ Hours or } 9 \text{ Hours } 30 \text{ Min}$$

List of components and their ratings.

- 1) PLC controller (1 no)
 - Power rating – 300 W
 - Voltage rating – 24 V
 - Current rating – 2 to 10 A
- 2) GSM + GPS module (1 no)
 - Power rating – 1.65 to 2.5 W
 - Voltage rating – 3.3 to 5 V
 - Current rating – 500 mA
- 3) NEMA 34 Stepper motor with drivers(4 nos)
 - Power rating – 360 W (for 4: $4 \times 360 = 1440 \text{ W}$)
 - Voltage rating – 60 V
 - Current rating – 5.9 A
- 4) Roller platform motors(7 nos)
 - Power rating – 26.76 W (for 7: $7 \times 26.76 = 187.32 \text{ W}$)
 - Voltage rating – 12 V
 - Current rating – 2.22 A
- 5) BLDC motor for screwing(4 nos)
 - Power rating – 15 W (for 4: $4 \times 15 = 60 \text{ W}$)
 - Voltage rating – 15 V

- Current rating – 0.95 A

6) RFID reader + sensors Let it be 100 W

Total load = sum of power consumed by all components
 Total load in watt = $300 + 2.5 + 1440 + 187.32 + 60 + 100$
 $= 2,090.5 \text{ W}$

CONCLUSION

The approach to providing efficient battery swapping services centered around a battery swapping mobile station is established. Also explains the functioning of the architecture of the services, and the functions and roles each entity plays are explained in detail. The mechanisms used in the mobile station are explained in the detail. We also discuss the components required and their role in the process. The mobile application is discussed in detail and how the operation in each section. For charging of the auxiliary batteries through the solar panels on the roof is established and the mathematical analysis of the solar array is done. Also, the energy consumption of the mobile station is estimated along with the run-time of it with one complete charge. But, the limitation of this approach is the efficiency is decreased due to factors such as road accessibility, and traffic, etc. The initial cost for the establishment of the essential entities is high. For a successful approach, the battery design, capacity, charge density has to be universally constant, or else the battery swapping mechanism has to be changed for different designs. But future research can give a solution to these limitations, and make the application of battery swapping come true.

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